

11.6 Testing single coefficients and groups of coefficients

The tests most often printed by regression software are instances of the general theory: a t test for each coefficient, an F test that all slopes vanish, and on request an F test for a group of coefficients. This section derives each from the preceding sections, explains what each one tests, and studies how a group test and the individual tests inside it can disagree, in both directions.

11.6.1 The t test

When the hypothesis consists of a single estimable function, $q = 1$, the F statistic is the square of a statistic with a sign, and the sign allows one-sided alternatives.

Theorem 11.6.1 — The t test for an estimable function

Assume Eq. 7.3.1 with $r < n$. Let $\lambda \in \mathcal{C}(\mathbf{X}^\top)$, $\lambda \neq \mathbf{0}$, let $d \in \mathbb{R}$, let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}^\top \mathbf{X}$, and put

$$T = \frac{\lambda^\top \hat{\beta} - d}{s \sqrt{\lambda^\top \mathbf{G} \lambda}}, \quad s^2 = \frac{\text{SSE}}{n - r}. \quad (11.6.1)$$

- (a) $T \sim t(n - r, \delta)$ with $\delta = (\lambda^\top \beta - d) / (\sigma \sqrt{\lambda^\top \mathbf{G} \lambda})$.
- (b) T^2 equals the statistic F_H of Eq. 11.3.1 for $H : \lambda^\top \beta = d$, and $\gamma = \delta^2$. The two-sided test that rejects when $|T| > t_{n-r, \alpha/2}$ is the F test of H , with the same p-value.
- (c) The one-sided test that rejects when $T > t_{n-r, \alpha}$ has size exactly α for the hypothesis $\lambda^\top \beta \leq d$: its rejection probability is at most α whenever $\lambda^\top \beta \leq d$, with equality when $\lambda^\top \beta = d$. Its power increases strictly with $\lambda^\top \beta$.

Proof.

- (a) is Corollary 7.5.2. (b) With $\mathbf{A} = \lambda$, $\mathbf{W} = \lambda^\top \mathbf{G} \lambda > 0$ is a scalar, so $F_H = (\lambda^\top \hat{\beta} - d)^2 / (\lambda^\top \mathbf{G} \lambda s^2) = T^2$, and Eq. 11.3.2 gives $\gamma = \delta^2$. Under H , $T \sim t(n - r)$ is symmetric about zero, so $\Pr\{|T| > t\} = \Pr\{T^2 > t^2\}$ for every $t > 0$. Hence $t_{n-r, \alpha/2}^2 = F_\alpha(1, n - r)$, the two rejection regions coincide, and the two-sided t p-value $\Pr\{|T| \geq |T_{\text{obs}}|\}$ equals the F p-value $\Pr\{T^2 \geq T_{\text{obs}}^2\}$. (c) By Exercise (A2), the probability that a $t(n - r, \delta)$ variable exceeds a fixed constant is strictly increasing in δ , and δ is an increasing function of $\lambda^\top \beta$. At $\delta = 0$ the probability is α .

■

For a single coefficient β_j in a full-rank model, Eq. 11.6.1 becomes the familiar $t_j = \hat{\beta}_j / \{s \sqrt{c_{jj}}\}$, with c_{jj} the j th diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$. By Corollary 9.2.3, t_j^2 is the F statistic for dropping column j from the model. The t test of $\beta_j = 0$ therefore compares the model with all regressors

to the model with all *except* $\mathbf{x}_j$. It asks whether $\mathbf{x}_j$ adds anything *given the others*. By the Frisch–Waugh–Lovell theorem (Theorem 6.6.2), what it measures is the association of $\mathbf{y}$ with the part of $\mathbf{x}_j$ orthogonal to the other columns. The same coefficient can be “significant” in one model and not in another, and there is no contradiction: the two tests answer different questions.

The one-sided test is appropriate only when the direction was specified before seeing the data, for example when theory says a coefficient cannot be negative, or when only an increase would lead to action. Choosing the direction after seeing the sign of the estimate doubles the actual size.

11.6.2 Tests and intervals are one computation

The two-sided test of $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$ at level α *accepts* exactly when $|\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d| \leq t_{n-r, \alpha/2} s \sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$. Collecting the values of d that are accepted gives the interval

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \pm t_{n-r, \alpha/2} s \sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}},$$

which is the $1 - \alpha$ confidence interval Eq. 7.5.1. Every interval of this kind is the set of hypothesized values that a test does not reject, and every test can be read off an interval: reject $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$ iff d lies outside. The interval reports more. It shows which values are ruled out, and so how precisely the data determine $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, which the verdict of a single test does not. For several functions at once, inverting the F test of Theorem 11.3.2 gives an ellipsoid. [Chapter 12](#) develops both and the duality between them.

11.6.3 The overall F test

The most common reduced model is the one with no regressors at all. Let $\mathbf{X}$ contain the intercept column $\mathbf{1}$ and have rank r . The reduced model $E(\mathbf{Y}) = \beta_0 \mathbf{1}$ has $\text{SSE}_0 = \text{SST} = \sum_i (y_i - \bar{y})^2$, and $\text{SST} - \text{SSE} = \text{SSR}$ is the regression sum of squares. By Theorem 11.1.1 and Theorem 9.5.2,

$$F = \frac{\text{SSR}/(r-1)}{\text{SSE}/(n-r)} = \frac{R^2/(r-1)}{(1-R^2)/(n-r)} \sim F(r-1, n-r, \gamma), \quad \gamma = \frac{\|(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{X}\boldsymbol{\beta}\|^2}{\sigma^2}.$$

The **overall F test** asks whether any linear combination of the regressors helps to predict the response. The answer is almost always yes in observational data with a sensible choice of regressors, and a significant overall F is correspondingly weak information. It does not show that any particular regressor matters, that the model fits well, or that it predicts usefully: with n large, a model with $R^2 = 0.02$ has a highly significant overall F . A nonsignificant overall F is more informative. It says that the data give no evidence that the regressors, taken together, relate to the response at all.

11.6.4 Groups of coefficients

Between one coefficient and all of them lies the test of a group: whether the terms of a factor, the powers of a polynomial beyond the linear, or a set of related regressors can be dropped together. The statistic is the extra sum of squares of the group, divided by its degrees of freedom and by s^2 (Theorem 11.1.1). How does it relate to the t statistics of its members?

Proposition 11.6.2 — The group statistic is the largest single statistic

Let $H : \Lambda^\top \beta = \mathbf{d}$ be testable with Λ of full column rank q , let F_H be its statistic Eq. 11.3.1, and for $\mathbf{a} \in \mathbb{R}^q$, $\mathbf{a} \neq \mathbf{0}$, let $T(\mathbf{a})$ be the t statistic Eq. 11.6.1 for the single function $\mathbf{a}^\top \Lambda^\top \beta = \mathbf{a}^\top \mathbf{d}$. Then

$$qF_H = \max_{\mathbf{a} \neq \mathbf{0}} T(\mathbf{a})^2,$$

and the maximum is attained at $\mathbf{a} \propto \mathbf{W}^{-1}(\Lambda^\top \hat{\beta} - \mathbf{d})$, where $\mathbf{W} = \Lambda^\top \mathbf{G} \Lambda$. In particular $t_j^2 \leq qF_H$ for each coefficient β_j in a tested group $\beta_j = 0$, $j \in J$, with $q = |J|$.

Proof. Write $\mathbf{u} = \Lambda^\top \hat{\beta} - \mathbf{d}$. The function $\mathbf{a}^\top \Lambda^\top \beta$ is estimable with $\lambda = \Lambda \mathbf{a} \neq \mathbf{0}$ (the columns of Λ are independent), and $\lambda^\top \mathbf{G} \lambda = \mathbf{a}^\top \mathbf{W} \mathbf{a}$, so $T(\mathbf{a})^2 = (\mathbf{a}^\top \mathbf{u})^2 / (s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a})$. The matrix $\mathbf{W}$ is positive definite (Theorem 9.2.4(a)) and has a positive definite square root (Theorem 1.7.6). By the Cauchy–Schwarz inequality (Proposition 1.1.1) applied to $\mathbf{W}^{1/2} \mathbf{a}$ and $\mathbf{W}^{-1/2} \mathbf{u}$,

$$(\mathbf{a}^\top \mathbf{u})^2 \leq (\mathbf{a}^\top \mathbf{W} \mathbf{a})(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}),$$

with equality iff $\mathbf{W}^{1/2} \mathbf{a}$ is proportional to $\mathbf{W}^{-1/2} \mathbf{u}$, that is, $\mathbf{a} \propto \mathbf{W}^{-1} \mathbf{u}$ (or $\mathbf{u} = \mathbf{0}$). Dividing by $s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a}$ gives $T(\mathbf{a})^2 \leq \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} / s^2 = qF_H$, with equality at the stated $\mathbf{a}$. For the coefficient β_j , take $\mathbf{a}$ to be the corresponding coordinate vector. ■

The proposition is the geometric heart of Scheffé’s method for multiple comparisons, and it explains the two ways in which a group test and its members can disagree.

Dilution. A single large t_j need not make the group significant. The group test rejects when $qF_H > qF_\alpha(q, n-r)$, and $qF_\alpha(q, n-r)$ grows with q (Exercise (B2)). A t_j^2 just above $F_\alpha(1, n-r)$ can therefore leave qF_H below the group’s critical value, especially when the other members of the group contribute nothing. This is what happened in Example (Do regions matter?): the Northeast contrast had $|t| = 2.519$, but the three-degree-of-freedom test of all regional differences was not significant.

Collinearity. Conversely, the group can be highly significant while every member has a small t . The direction $\mathbf{a}$ that attains the maximum is generally not a coordinate direction. When the estimates are strongly correlated, the data can say clearly that *some* combination of the coefficients is nonzero without being able to attribute it to any single one.

Example. (Five regressors for the murder rate)

Regress the murder rate of the 50 states on poverty, high-school graduation, single-parent households, percentage white and urbanization, with an intercept ($p = 6$, $n - p = 44$).

Regressor	Estimate	Standard error	t	p-value
intercept	−10.0586	17.6785	−0.569	0.572
poverty	0.2378	0.1412	1.684	0.099
high-school graduates	−0.0047	0.1530	−0.030	0.976

single-parent households	0.4280	0.1372	3.120	0.003
white	0.0108	0.0270	0.398	0.692
urban	0.0046	0.0145	0.315	0.754

The overall test gives $R^2 = 0.643$ and $F = 15.87$ on 5 and 44 degrees of freedom, $p = 6.4 \times 10^{-9}$. Only single parenthood has a t statistic beyond the critical value 2.015. The group of three demographic and educational regressors, high-school graduation, percentage white and urbanization, has $F = 0.096$ on 3 and 44 degrees of freedom, $p = 0.962$: the data give no evidence that these three add anything once poverty and single parenthood are in the model. Every t_j^2 of the group is at most $3F = 0.29$, as [Proposition 11.6.2](#) requires.

This group was chosen here because its members have small t statistics. A group selected that way will tend to have a small F , and its large p-value makes dropping the group look safer than it is. Groups should be defined by the scientific question, before the individual results are seen.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statelcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
names = ["poverty", "hs_grad", "single", "white", "urban"]
y = data["murder"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), data[names]])
p = X.shape[1]

C = np.linalg.inv(X.T @ X)
b = C @ X.T @ y
sse = np.sum((y - X @ b) ** 2)
s2 = sse / (n - p)
se = np.sqrt(s2 * np.diag(C))
t = b / se
pv = 2 * stats.t.sf(np.abs(t), n - p)
for name, bj, sj, tj, pj in zip(["intercept"] + names, b, se, t, pv):
    print(f"{name:10s} {bj:9.4f} {sj:8.4f} {tj:7.3f} {pj:7.4f}")
```

[Run this code online](#)

```
sst = np.sum((y - y.mean()) ** 2)
R2 = 1 - sse / sst
F_all = (R2 / (p - 1)) / ((1 - R2) / (n - p))

def extra_F(keep):
    """F test that the coefficients not in `keep` (columns of X) are zero."""
    Xr = X[:, keep]
    br, *_ = np.linalg.lstsq(Xr, y, rcond=None)
    sse_r = np.sum((y - Xr @ br) ** 2)
    q = p - len(keep)
    F = ((sse_r - sse) / q) / s2
    return F, stats.f.sf(F, q, n - p)
```

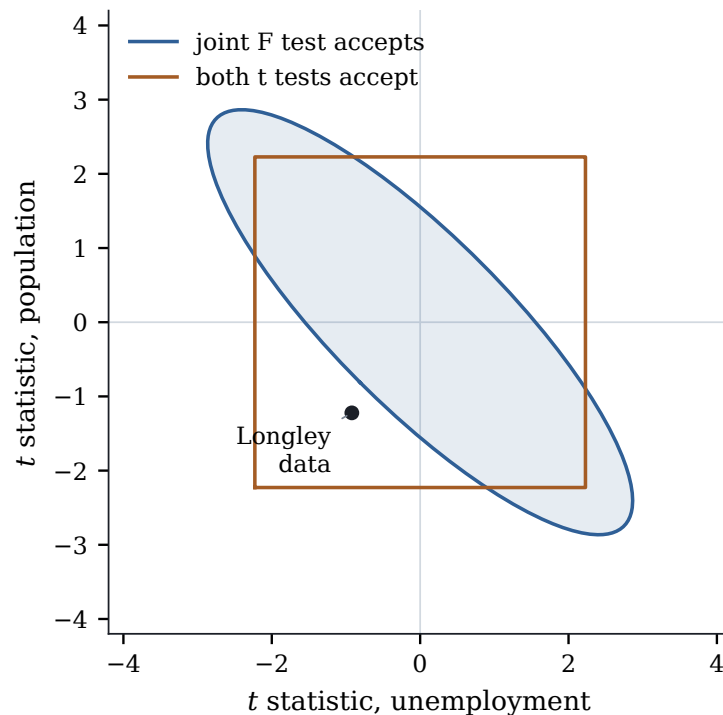


Figure 11.6.1: Acceptance regions for the hypothesis that the unemployment and population coefficients in Longley's regression are both zero, drawn in the plane of the two t statistics. The square accepts when both individual t tests accept at level 0.05. The ellipse accepts when the joint F test does. The black point is the data.

```
F_group, p_group = extra_F([0, 1, 3]) # drop hs_grad, white, urban
print(f"overall: R2 = {R2:.3f}, F = {F_all:.2f} on ({p - 1}, {n - p})")
print(f"hs_grad, white, urban: F = {F_group:.3f}, p = {p_group:.3f}")
```

[Run this code online](#)

Example. (Jointly significant, individually not)

Longley's data record US employment for the 16 years 1947 to 1962, with five economic series: the GNP deflator, GNP, unemployment, the size of the armed forces and population. Regress employment on the five series and an intercept, leaving out the time trend, so that $n - p = 10$. For unemployment and population, the t statistics are -0.921 and -1.222 , with p -values 0.379 and 0.250 . Neither comes near the critical value 2.228 . The test that *both* coefficients are zero gives $F = 7.175$ on 2 and 10 degrees of freedom, $p = 0.0117$.

Figure 11.6.1 shows why. The two estimates have correlation -0.840 . Under the hypothesis, their t statistics are then likely to have opposite signs, and the F test's acceptance region is an ellipse stretched along the line $t_1 = -t_2$. The observed pair has both statistics negative, a combination the ellipse excludes although each coordinate on its own is unremarkable. The data are clear that unemployment and population *together* matter, and unclear about how to divide the credit. With 16 consecutive years of trending series, the errors are also unlikely to be independent, a caution taken up in Chapter 21.

```
lg = sm.datasets.longley.load_pandas()
```

```

Z = sm.add_constant(lg.exog.drop(columns="YEAR"))
res = sm.OLS(lg.endog, Z).fit()
joint = res.f_test("UNEMP = 0, POP = 0")
print(res.tvalues[["UNEMP", "POP"]].round(3).to_dict(), res.pvalues[["UNEMP", "POP"]].round(3).to_dict())
print(f"joint F = {float(joint.fvalue):.3f}, p = {float(joint.pvalue):.4f}")

```

[Run this code online](#)

The figure also shows the opposite disagreement. Points inside the ellipse but outside the square have one significant t and a nonsignificant F . Neither verdict is wrong: the individual tests ask about the coordinate directions, and the joint test asks about all directions at once, paying for that breadth with a larger critical value.

Warning (Several t tests are not a test of the group). Rejecting the group hypothesis “all $\beta_j = 0$, $j \in J$ ” whenever at least one $|t_j|$ exceeds $t_{n-r, \alpha/2}$ is not a level- α test: when the k coefficient estimates are independent and $n - r$ is large, its size is about $1 - (1 - \alpha)^k$, about 0.23 for $k = 5$ at $\alpha = 0.05$. (The t statistics share the estimate s , so they are not exactly independent.) Reporting the smallest of several p-values as if it were the only one overstates the evidence in the same way. The F test of the group, or a multiple-comparison procedure designed for the family (Chapter 13), controls the familywise error rate. Choosing regressors by their t statistics and then testing the chosen model on the same data invalidates all the p-values of the final fit. Chapter 29 returns to this problem.

11.6.5 Exercises

11.6.5.1 A. Check your understanding

Exercise A1.

Using the table of [Example \(Five regressors for the murder rate\)](#), compute the F statistic for dropping poverty from the five-regressor model and its p-value, without refitting.

Exercise A2.

Show that if the direction of a one-sided t test is chosen after seeing the sign of T , and the test then rejects when $|T| > t_{n-r, \alpha}$, its size is 2α .

11.6.5.2 B. Practice

Exercise B1.

In a full-rank model, show that the t statistic for $\beta_j = \beta_k$ is

$$T = \frac{\hat{\beta}_j - \hat{\beta}_k}{s \sqrt{c_{jj} + c_{kk} - 2c_{jk}}},$$

where c_{jk} are the entries of $(\mathbf{X}^\top \mathbf{X})^{-1}$. Show that it equals the t statistic of the coefficient

of $\mathbf{x}_j$ in the reparameterized model with regressors $\mathbf{x}_j$ and $\mathbf{x}_j + \mathbf{x}_k$ in place of $\mathbf{x}_j$ and $\mathbf{x}_k$.

Exercise B2.

Show that $qF_\alpha(q, \nu)$ is strictly increasing in q for fixed ν and α . Deduce that a single t statistic with t^2 just above $F_\alpha(1, \nu)$ does not force the rejection of a group containing it.

Solution. $qF(q, \nu)$ has the law of $U_q/(V/\nu)$ with $U_q \sim \chi^2(q)$ independent of $V \sim \chi^2(\nu)$. Taking $U_{q+1} = U_q + W$ with $W \sim \chi^2(1)$ independent of U_q and V , we have $U_{q+1}/(V/\nu) > U_q/(V/\nu)$ with probability one, so $\Pr\{U_{q+1}/(V/\nu) > c\} > \Pr\{U_q/(V/\nu) > c\}$ for every $c > 0$ (strictly, since $\Pr\{U_q/(V/\nu) \leq c < U_{q+1}/(V/\nu)\} > 0$). Taking $c = qF_\alpha(q, \nu)$, where the right side equals α , shows that the upper α point of $U_{q+1}/(V/\nu)$ exceeds $qF_\alpha(q, \nu)$. For the deduction, if t_j^2 lies between $F_\alpha(1, \nu)$ and $qF_\alpha(q, \nu)$ and the other members contribute nothing, then qF_H can equal $t_j^2 < qF_\alpha(q, \nu)$, and the group is not rejected.

Exercise B3.

For two coefficients with t statistics t_1, t_2 whose estimates have correlation ρ , show that the F statistic for both being zero is

$$F = \frac{t_1^2 + t_2^2 - 2\rho t_1 t_2}{2(1 - \rho^2)}.$$

Check the value 7.175 of [Example \(Jointly significant, individually not\)](#) from $t_1 = -0.921$, $t_2 = -1.222$ and $\rho = -0.840$, and find the value of F had the correlation been $+0.840$.

Solution. Let $\mathbf{D} = \text{diag}(\text{se}_1, \text{se}_2)$ and $\mathbf{R}$ the correlation matrix of the estimates, so that the estimated covariance is $\mathbf{DRD}$. With $\mathbf{A}$ selecting the two coefficients, $2F = \hat{\boldsymbol{\beta}}_J^\top (\mathbf{DRD})^{-1} \hat{\boldsymbol{\beta}}_J = \mathbf{t}^\top \mathbf{R}^{-1} \mathbf{t}$, where $\mathbf{t} = \mathbf{D}^{-1} \hat{\boldsymbol{\beta}}_J$, and $\mathbf{R}^{-1} = (1 - \rho^2)^{-1} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$. Numerically, $t_1^2 + t_2^2 = 2.341$, $-2\rho t_1 t_2 = 1.890$ and $2(1 - \rho^2) = 0.589$, so $F = 7.175$, as in the example, up to the rounding of these three terms. With $\rho = +0.840$ the middle term changes sign and $F = 0.765$, about $(2.341 - 1.890)/0.589$: the same two t statistics would then be entirely unremarkable.

11.6.5.3 C. Going deeper

Exercise C1.

Show that $\max T(\boldsymbol{\lambda})^2$ over all nonzero estimable $\boldsymbol{\lambda}$, for the hypotheses $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = 0$, equals rF_* , where F_* is the statistic for the reduced model $E(\mathbf{Y}) = \mathbf{0}$. Deduce that $\Pr\{T(\boldsymbol{\lambda})^2 \leq rF_\alpha(r, n - r) \text{ for all estimable } \boldsymbol{\lambda}\} = 1 - \alpha$ when $\boldsymbol{\beta} = \mathbf{0}$. What changes if $\boldsymbol{\beta} \neq \mathbf{0}$ and $T(\boldsymbol{\lambda})$ is centred at $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$? ([Chapter 13](#) builds simultaneous inference on this.)

Exercise C2.

Find the likelihood ratio test of $\boldsymbol{\lambda}^\top \boldsymbol{\beta} \leq d$ against $\boldsymbol{\lambda}^\top \boldsymbol{\beta} > d$ for estimable $\boldsymbol{\lambda}$, and show that it rejects for large values of T in Eq. 11.6.1. (Maximize the likelihood under the inequality constraint: if $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \leq d$ the unrestricted maximum is feasible; otherwise the maximum is on the boundary.)

Solution. If $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \leq d$, the unrestricted maximizer satisfies the constraint and $\Lambda = 1$. Otherwise the log-likelihood, maximized over σ^2 , is a decreasing function of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$, which is a convex quadratic in $\mathbf{b}$ with minimum at $\hat{\boldsymbol{\beta}}$. Its minimum over the half-space $\boldsymbol{\lambda}^\top \mathbf{b} \leq d$ therefore lies on the boundary $\boldsymbol{\lambda}^\top \mathbf{b} = d$, where it equals $\text{SSE} + (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d)^2 / \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ (Eq. 7.3.3). So $\Lambda = \{1 + T^2 / (n - r)\}^{-n/2}$ when $T > 0$ and $\Lambda = 1$ when $T \leq 0$. This is a nonincreasing function of T , strictly decreasing for $T > 0$, so $\Lambda \leq k < 1$ iff $T \geq c$ for some $c > 0$, which is the one-sided t test.